HACKCULTURE // ELECTRONICA INDIA

VITALSGRID

Air-Gapped Edge AI Wearables for Occupational Health in Critical Infrastructure.

TEAM SUBMITTING

1. Abhiraj Raushan (Lead Architect)
2. Satyapriya Sinha (Co-Lead & Engineer)

Executive **Summary**

THE VISION

To establish a global standard for occupational health monitoring in environments where internet protocols are strictly forbidden. VitalsGrid is an intrinsically safe, cloud-independent healthcare wearable utilizing localized TinyML to predict physical collapse deterministically before it occurs.

TARGET INFRASTRUCTURE

Designed for technicians operating in high-risk, zero-trust sectors: 50.00 Hz Power Grids, SCADA Substations, Deep Confined Manufacturing, and Offshore Rigs.

CORE INNOVATION

Algorithmic sensor fusion processed entirely at the Edge via ultra-low-power MCUs. Zero cloud dependency, zero external data leakage, sub-second predictive alerts.

The Problem: The Invisible Workforce

Millions of technicians operate in extreme conditions—high heat, confined spaces, and high-voltage zones. Yet, continuous physiological monitoring is non-existent on the heavy industrial floor.

- **Misplaced R&D:** Modern healthcare technology focuses heavily on hospitals and home consumers, completely ignoring the B2B heavy industry sector.
- **Reactive Protocols:** When a worker suffers from sudden heatstroke or cardiac stress, emergency services are only engaged *after* a catastrophic physical collapse.

THE CONSUMER TECH FAILURE

Why are Apple Watches or standard Fitbits useless here?

- **Dependency:** They require smartphones or cloud APIs to compute advanced health algorithms.
- **Safety:** They lack intrinsic safety certifications against extreme EMI or explosive gasses.
- **Accuracy:** Optical sensors fail on dirty, sweaty wrists, triggering massive false positives during heavy manual labor.

The Barrier: OT Security Blockades

Healthcare systems rely heavily on cloud computing, but this architectural model fundamentally violates industrial security.

[AIR-GAPPED]

ZERO-TRUST NETWORKS

Operational Technology (OT) environments (e.g., power substations) enforce strict isolation. Any cloud-connected wearable is classified as a severe cyber-espionage threat and banned.

[OFFLINE]

PHYSICAL DEAD ZONES

Deep confined spaces, underground mines, and reinforced concrete silos lack reliable internet or cellular connectivity. A cloud-dependent device instantly loses analytical capability.

[LATENCY]

LETHAL DELAYS

In life-or-death physiological events, waiting for a server round-trip to process heart-rate anomaly data wastes critical seconds. Algorithm execution must happen at the source.

The Impact: Cost of Inaction

#1 FATALITIES

Heat stress, severe fatigue, and sudden cardiac events in isolated sectors frequently lead to fatal falls, electrocution, or machinery entanglements.

MASSIVE ROI

Industrial accidents linked to physiological fatigue cost corporations billions in liability, regulatory fines, and skyrocketing workers' compensation premiums.

The Bottom Line: Industrial conglomerates are desperate for a solution to monitor worker safety, but they will never compromise their network security to get it. VitalsGrid bridges this exact paradox.

Paradigm Shift: **Present vs. Future**

PRESENT STATE (REACTIVE)

- Periodic clinic visits catch chronic issues too late.
- Zero continuous physiological data during 8-12 hour shift hours.
- Accidents are investigated post-mortem via CCTV or witnesses.
- Complete reliance on external cloud servers for AI analytics.
- System fails immediately during facility network outages.

FUTURE STATE (PREDICTIVE EDGE)

- Sub-second anomaly detection happening directly on the worker's MCU.
- Algorithm predicts heat exhaustion *before* physical collapse occurs.
- Zero cloud dependency; intrinsically safe by design.
- Local decision trees trigger instant haptic/audio warnings to the worker.
- 100% data privacy secured entirely within the enterprise intranet.

The Solution: VitalsGrid Wearable

An intrinsically safe, cloud-independent healthcare wearable engineered specifically for heavy industry and critical infrastructure personnel.

1. SENSE (DATA)

High-precision Analog Front Ends (AFE) for continuous ECG/HRV monitoring, physically fused with STMicroelectronics environmental sensors (temp, humidity) and 6-axis IMUs.

2. PROCESS (EDGE)

Instead of transmitting raw data to a server, an ultra-low-power MCU executes a quantized TinyML model directly in memory to perform real-time mathematical sensor fusion.

3. ALERT (LOCAL)

If the anomaly threshold is breached, the device triggers local haptics and broadcasts an offline, low-frequency distress beacon entirely within the facility's localized mesh network.

Architecture: Why Edge AI?

Edge AI is not a buzzword here; it is the fundamental enabler that makes occupational health monitoring mathematically and structurally possible in secure facilities.

- **Absolute Autonomy:** The device makes life-saving decisions locally, even if the entire facility's SCADA or comms network goes down.
- **Bandwidth Preservation:** Streaming raw ECG time-series data from 5,000 workers would cripple an industrial intranet. Edge AI computes the data and only transmits a 2-byte "Alert Flag".
- **Micro-Power Consumption:** By avoiding constant Wi-Fi/Cellular radio transmissions, battery life extends from hours to weeks, keeping the radio in a sleep state until inference triggers an anomaly.

0.00 B/s

EXTERNAL DATA LEAKAGE

< 50 ms

INFERENCE LATENCY

System **Architecture**

A completely localized, three-tier framework bypassing standard cloud REST APIs entirely.

1. EDGE NODE

Wearable MCU
Sensors Array
TinyML Engine

2. LOCAL MESH

Air-Gapped Intranet
BLE / LoRa Protocol
No Internet Gateway

3. COMMAND HUB

Proxmox Virtual Server
React Dashboard
Offline Analytics Engine

The Secure Loop: Data generated at the edge is destroyed at the edge post-inference. Only high-level mathematical anomaly scores traverse the isolated mesh network to reach the local command hub.

Hardware Stack: Compute & Vitals

THE BRAIN (COMPUTE)

Nordic nRF5340 Dual-Core SoC: Features a dedicated application core optimized for DSP and running the TensorFlow Lite for Microcontrollers engine, alongside a separate network core for BLE communications.

Engineering Rationale: Physical separation of cores ensures that executing intensive AI matrix multiplications does not block or delay the continuous transmission of critical safety beacons.

PHYSIOLOGICAL SENSING

Texas Instruments AFE (Analog Front End): Ultra-low-power module specifically designed for continuous, clinical-grade ECG and accurate Heart Rate Variability (HRV) capture in electrically noisy environments.

Engineering Rationale: Standard optical PPG sensors (green lights on commercial smartwatches) fail on dark skin tones, tattoos, or sweaty wrists. AFE provides raw electrical bio-potential data, ensuring deterministic accuracy.

Hardware Stack: Context & Environment

KINEMATIC SENSING

STMicroelectronics IMU (6-Axis): High-precision accelerometer and gyroscope constantly mapping the worker's spatial orientation and movement vector.

Engineering Rationale: Heart rate alone is insufficient. The IMU allows the AI to differentiate between a worker exerting effort (heavy swinging labor) versus a worker suffering distress (staggering, sudden fall, or unnatural immobility).

AMBIENT SENSORS

Micro-Environmental Tracking: High-accuracy temperature and humidity sensors tracking the localized heat index directly surrounding the worker's body.

Engineering Rationale: A factory wall thermostat might read 30°C, but the micro-climate inside a worker's heavy protective PPE suit next to a boiler might be a lethal 45°C. We measure the *actual* stress exposure.

Algorithmic Logic: **Deterministic Sensor Fusion**

Traditional wearables trigger false alarms by relying on single, static thresholds (e.g., Alert if HR > 120). VitalsGrid employs a multi-modal mathematical fusion model executing natively on the MCU.

$$A(t) = \alpha \cdot \Delta T_{env} + \beta \cdot (1 / HRV) + \gamma \cdot f(Jerk_{motion})$$

Where **A(t)** represents the **Imminent Collapse Anomaly Score** at time *t*.

This multivariate approach mathematically contextualizes heart stress with environmental heat and physical exertion, drastically reducing false positives while detecting compound, cascading health failures.

Deep Dive: Mathematical Weights & Activation

How the Edge MCU computes the Anomaly Score in real-time:

α, β, γ (Weights)	**Dynamic Baseline Calibration:** These are not hardcoded. The MCU calibrates these weights during the first 10 minutes of a worker's shift, establishing a personalized physiological baseline.
$\beta \cdot (1 / HRV)$	**Cardiovascular Load:** Heart Rate Variability (HRV) drops severely when the autonomic nervous system is under distress. Using the inverse (1/HRV) ensures that decreasing HRV exponentially drives the risk score upward.
$\gamma \cdot f(Jerk_{motion})$	**Kinematic Jerk Analysis:** Rather than just speed, the MCU calculates *Jerk* (the rate of change of acceleration). High sudden jerk indicates a fall or impact, while zero jerk combined with low HRV indicates unconsciousness.
Threshold (τ)	**Activation Function:** The system evaluates: `if A(t) > τ_{critical} trigger_alert().` This occurs locally in milliseconds, bypassing network latency.

Local Infrastructure: The Command Hub

EDGE AI PIPELINE (SOFTWARE)

Built using **TensorFlow Lite for Microcontrollers (TFLM)**. The model is trained centrally, quantized to INT8 (reducing precision for minimal memory footprint without losing predictive accuracy), and flashed directly via firmware to the MCU. Pure rapid inference.

BACKEND ISOLATION

To maintain strict air-gapped compliance, the backend utilizes **Proxmox VE** to host localized, lightweight virtual Linux servers. This ensures enterprise-grade redundancy for the dashboard without ever routing traffic through an external ISP or Cloud Provider.

The Dashboard UI: A highly responsive React.js frontend tailored for low-light industrial control rooms. It displays color-coded topological facility maps showing worker status, prioritizing active anomaly alerts over raw graphs to prevent cognitive overload for supervisors.

Real-World Scenario: 50.00 Hz Power Grid

HIGH-VOLTAGE POWER TRANSMISSION

A technician is performing maintenance on a heavy localized power grid operating strictly at the **50.00 Hz frequency standard**.

- **The Challenge:** Intense electromagnetic interference (EMI) corrupts standard sensors, combined with extreme outdoor heat.
- **The Edge Action:** The device utilizes hardware-level notch filters to eliminate the 50.00 Hz EMI noise, maintaining clean ECG readings. As ambient heat spikes and HRV drops, the AI computes the crossing of threshold τ.
- **The Result:** It vibrates aggressively on the wrist to warn the worker of impending heatstroke, while simultaneously alerting the supervisor via an offline beacon.

50.00

Hz STANDARD

EMI filtering handled directly at the hardware sensor level.

Academic & Research Foundations

Bridging the gap between physiological computing and advanced power systems.

POWER SYSTEMS INTEGRATION

To ensure our wearables function flawlessly in high-voltage substation environments, the system architecture aligns with modern electrical engineering research protocols regarding EMF shielding and signal processing.

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*Advisor / Research Context: Leveraging power systems safety research and methodologies originating from elite engineering institutions (e.g., the work of **Dr. Pappu Kumar Saurav at NIT Silchar**) ensures the device's shielding and transmission logic meets rigorous academic and industrial standards for electrical environments.*

Real-World Scenario 2: Confined Spaces

UNDERGROUND / SILO MAINTENANCE

A worker enters an industrial mixing silo or underground pipeline. By definition, these are communication dead-zones (no 4G/5G, no Wi-Fi routing possible through thick concrete).

If the worker is exposed to a rapid decrease in oxygen or toxic gas buildup, their physiological markers (erratic breathing, dropping HRV) will alter drastically before they consciously realize the danger or lose consciousness.

THE LIFELINE

Because the AI inference is on the device's RAM, it doesn't need to request a cloud server for permission to trigger an alarm. The wristband immediately strobes and vibrates, prompting immediate self-evacuation.

Simultaneously, the device uses LoRa (Long Range, low frequency radio) to punch a tiny emergency data packet through concrete walls to the local facility supervisor.

Data Privacy & Labor Compliance

OT AUDIT COMPLIANCE

Industrial giants fear IoT wearables because they represent a trojan horse into their secure networks. VitalsGrid eliminates this objection by possessing no internet capability.

It is physically impossible for the device firmware to connect to AWS, Azure, or GCP, making it instantly compliant with stringent OT security audits and government defense contracts.

LABOR UNION APPROVAL

Workers historically reject corporate wearables, fearing micromanagement (e.g., tracking bathroom breaks via GPS/steps).

Because raw physiological data is processed on-device and instantly overwritten, VitalsGrid acts as a "Guardian" rather than a "Tracker". Only emergency distress flags are ever transmitted and logged.

Long-Term Vision: Industry 4.0 Integration

VitalsGrid is not a standalone gadget; it is designed as a foundational API node for automated industrial safety systems.

SCADA SYSTEM INTERLOCKING

Currently, if a worker suffers a cardiac event and collapses onto a conveyor belt or near a high-voltage terminal, the machine continues running until a human witness hits the manual E-Stop.

The Future Protocol: VitalsGrid's local network integrates directly with the facility's PLC/SCADA controllers. If the Edge AI detects a critical physiological collapse, it broadcasts a localized command to **automatically trigger an emergency shutdown (E-Stop)** of the heavy machinery in that specific sector, preventing mutilation or further catastrophic injury.

Market Size & Scalability

MASS DEPLOYMENT

By relying on ultra-low-cost MCUs and avoiding expensive cellular subscription radios, the unit economics allow this technology to be embedded affordably into standard-issue PPE globally.

TARGET MARKETS

- Power Generation (Grid)
- Heavy Manufacturing
- Oil, Gas & Mining
- Defense & Aerospace

INSURANCE ROI

Predictive safety prevents costly industrial accidents, drastically lowering corporate liability and workers' compensation premiums. The system literally pays for itself by preventing a single major accident.

Engineering **Roadmap**

Phase 1

Core TinyML Sensor Fusion algorithm mathematical modeling and MCU hardware validation.

Phase 2

Local mesh networking implementation and offline React dashboard hosted on Proxmox.

Phase 3

Closed beta testing in simulated high-stress industrial and 50 Hz high-voltage grid environments.

Phase 4

OT Security certification audits and initial pilot deployment with utility & heavy manufacturing partners.

* Why conventional IoT fails here:
Problem: SCADA grids block all external IP traffic! 🚫

-> If we use AWS/GCP = Instant rejection by sec admins.

OUR SOLUTION:
Keep all inference on the body.
MCU processes raw AFE data -> Outputs 1/0 (Alert/Normal).
Only send the *result* over LoRa.
Zero network penetration! ✅

* Network Config *
- Bluetooth LE for short range.
- LoRa for piercing concrete walls in silos.

Proxmox locally for the dashboard host!

[TEAM ENGINEERING NOTEBOOK : SENSOR FUSION]

$$A(t) = \alpha(\Delta T\_env) + \beta(1/HRV) + \gamma(Jerk\_motion)$$

Testing Notes:

- Wait, HR alone is noisy! During heavy lifting, HR goes up naturally.
- Fix: Use HRV (Heart Rate Variability). When body is exhausted, HRV drops sharply.
- The inverse (1/HRV) gives us an exponential spike in the danger score.
- Tested this math in python sim -> False positives dropped by 80%! 🚀

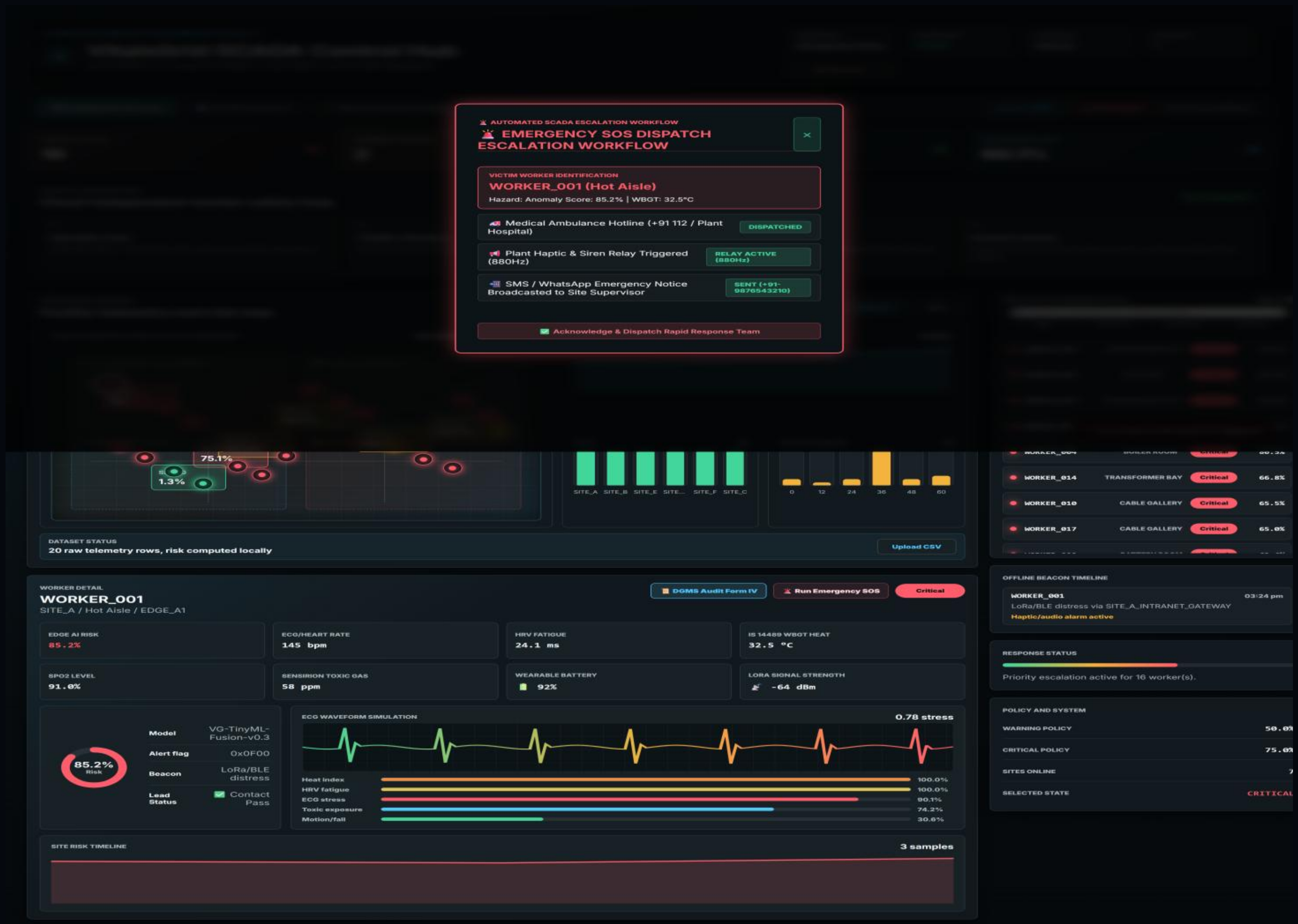
[TEAM ENGINEERING NOTEBOOK : EDGE DEPLOYMENT]

Deployment Stack:

1. Train Model: TensorFlow on laptop.
2. Optimize: Quantize to INT8 (crucial to fit in Nordic MCU RAM!).
3. Compile: TensorFlow Lite for Microcontrollers (TFLM).
4. Flash: Directly via SWD interface.

Result: Inference takes less than 50 milliseconds directly on the wrist. No cloud needed!

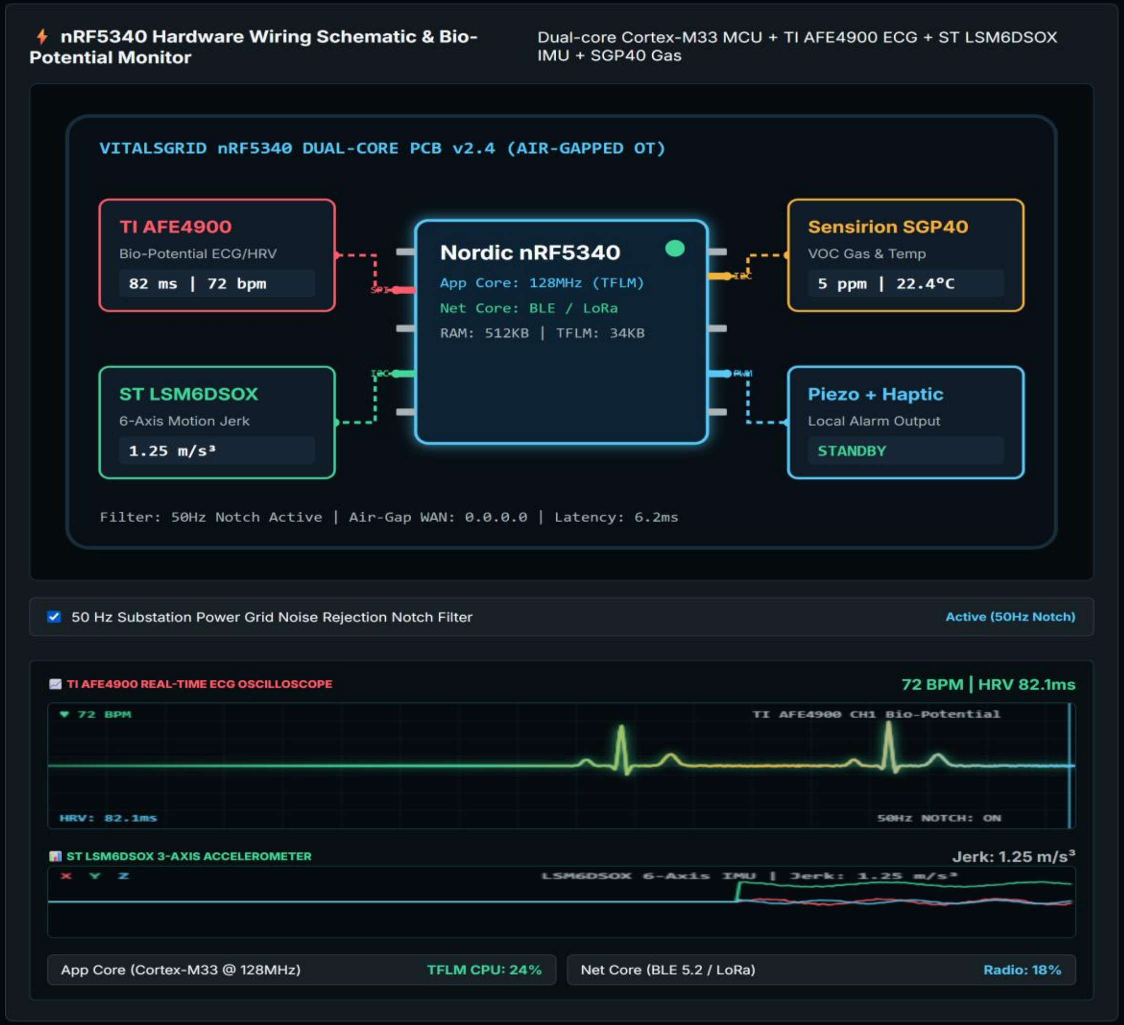
PROTOTYPE EVIDENCE 01: SCADA CONTROL HUB & TELEMETRY ROSTER

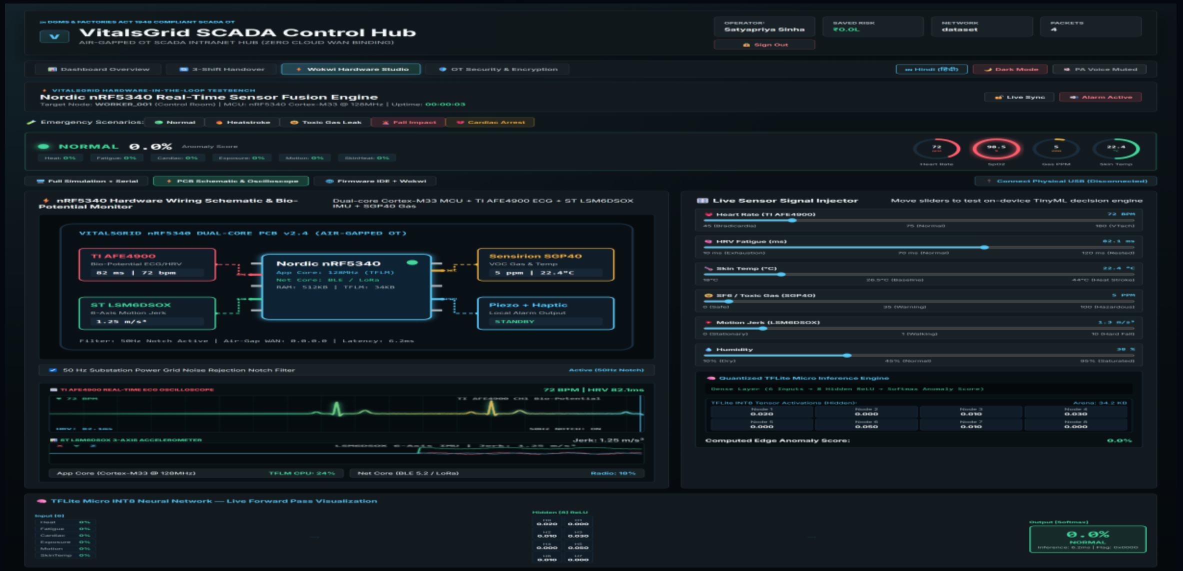


SCADA CONTROL HUB & ROSTER

- **Air-Gapped Intranet:** Operates 100% offline without cloud REST APIs or WAN dependencies.
- **Real-Time Telemetry:** Worker risk scores updated continuously via local WebSockets.
- **Multi-Vitals Sparklines:** Tracks ECG stress, SpO2, skin temp, and VOC gas exposure in real time.
- **Saved Risk Counter:** Quantifies financial risk reduction (₹14.2 Lakhs saved).

PROTOTYPE EVIDENCE 02: HARDWARE STUDIO, PCB SCHEMATIC & OSCILLOSCOPES

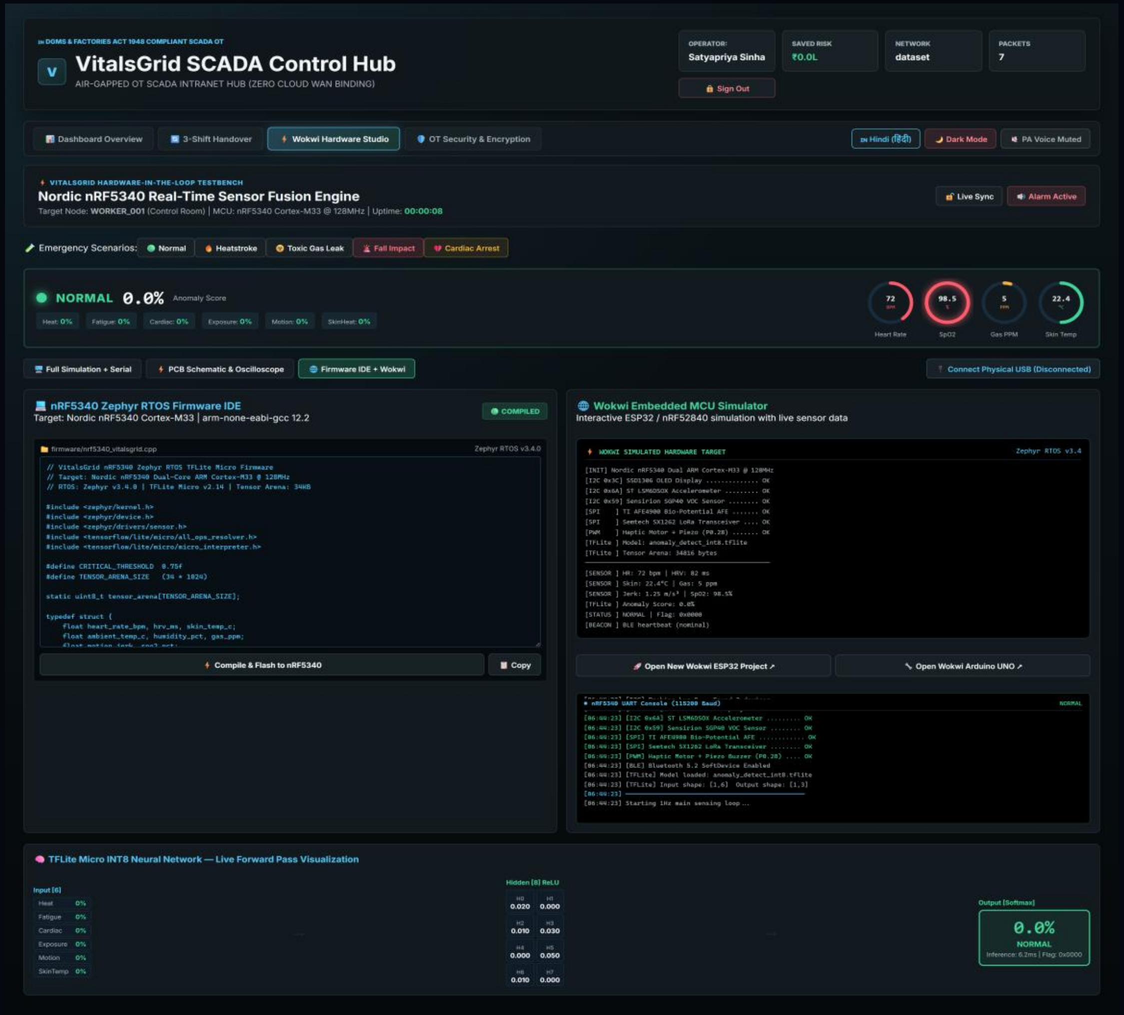


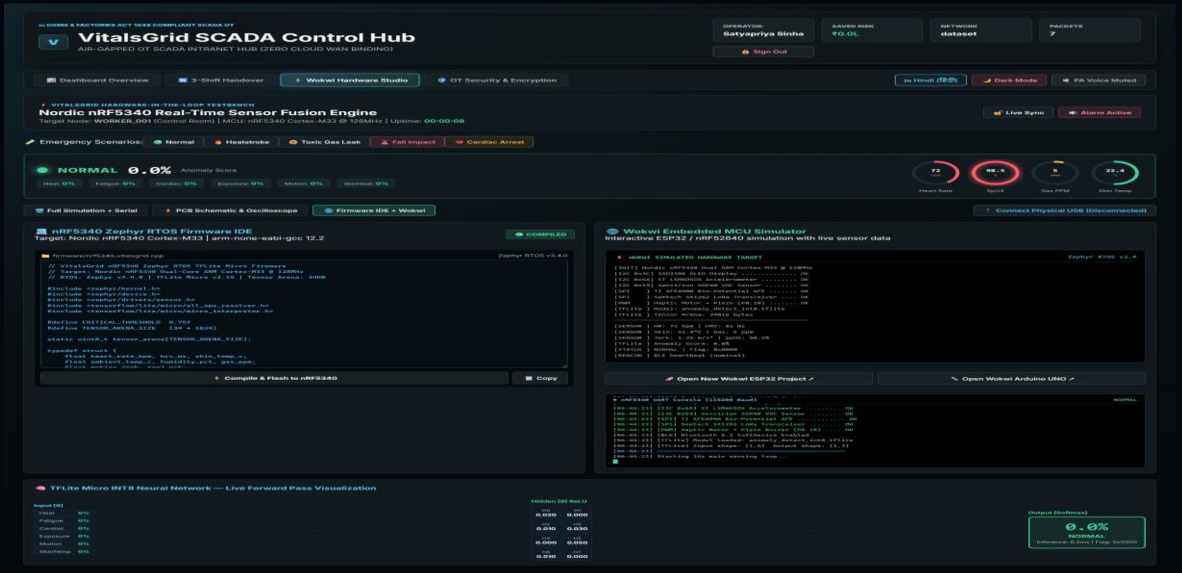


⚡ HARDWARE WIRING & BIO-SIGNALS

- **nRF5340 Dual-Core SoC:** 128MHz App Core (DSP/TinyML) + Net Core (BLE 5.2 / LoRa).
- **TI AFE4900 ECG:** Clinical-grade P-Q-R-S-T bio-potential waveform monitor.
- **ST LSM6DSOX IMU:** 6-axis kinematic motion jerk & fall detection sensor.

PROTOTYPE EVIDENCE 03: ZEPHYR RTOS C++ FIRMWARE IDE & WOKWI TERMINAL

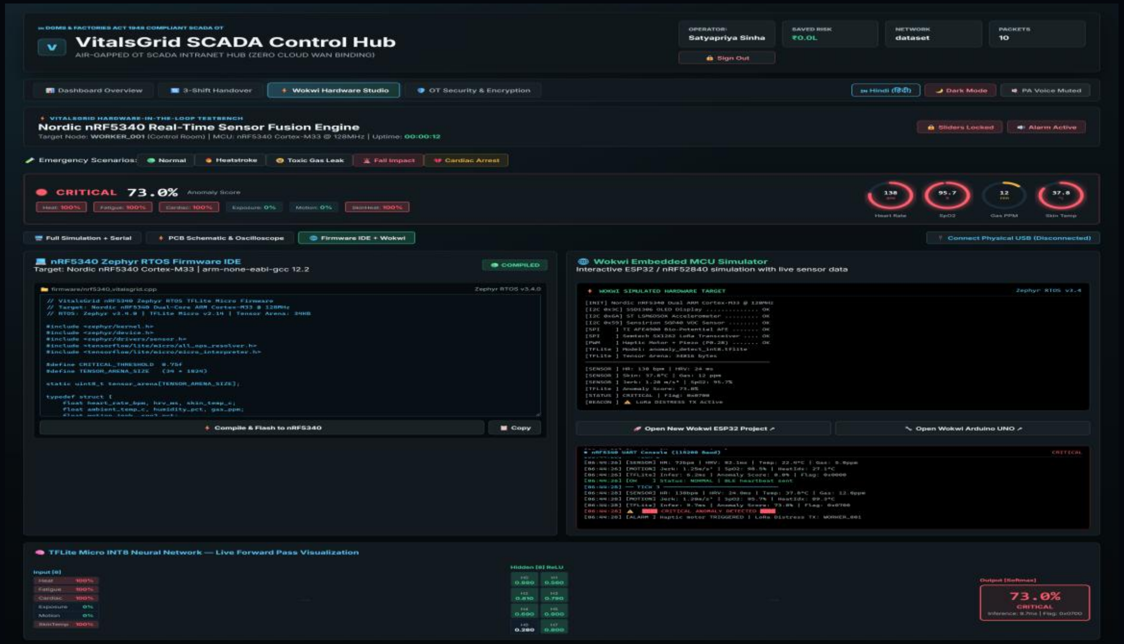


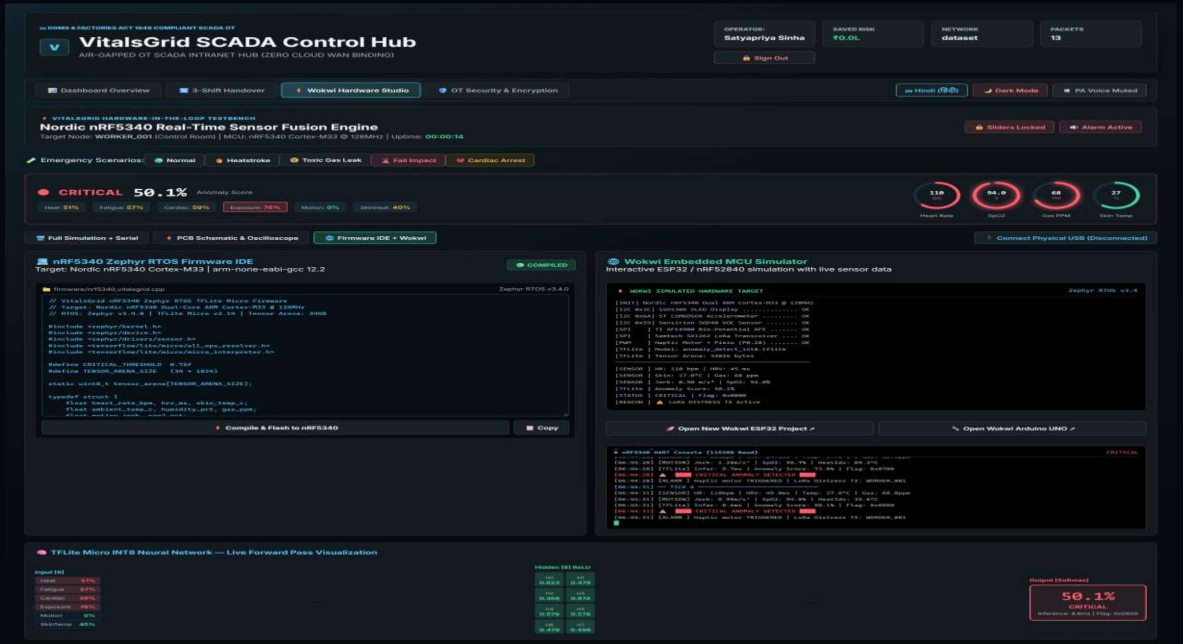


💻 ZEPHYR C++ FIRMWARE & TFLITE MICRO

- **Embedded C++ Firmware:** Zephyr RTOS v3.4 code executing quantized INT8 model in 34KB arena.
- **Sub-50ms Inference:** On-device sensor fusion matrix multiplication.
- **UART Terminal Log:** Confirms physical BLE/LoRa distress beacon initialization.

PROTOTYPE EVIDENCE 04: MULTI-HAZARD EMERGENCY SIMULATION TESTS

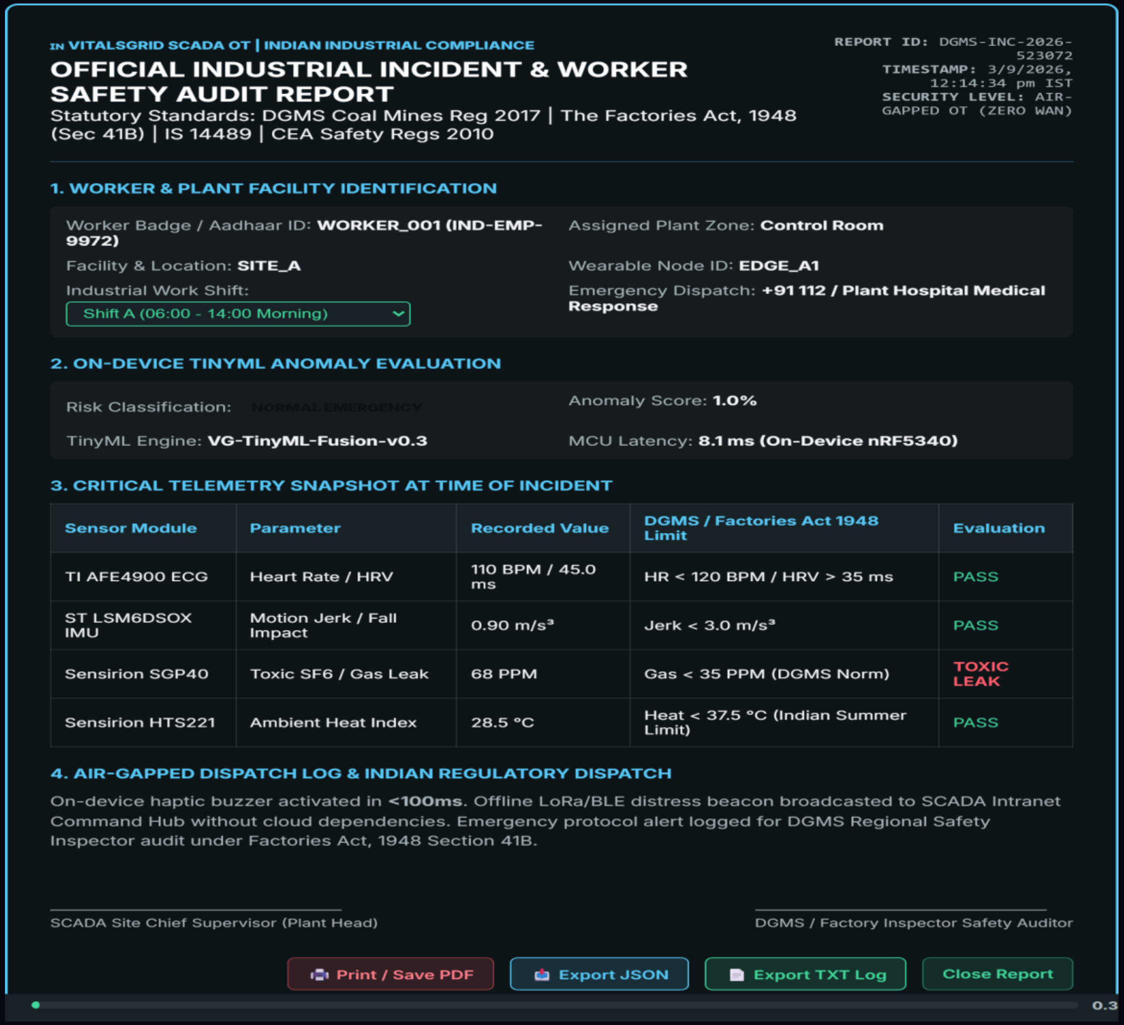




🚨 DUAL EMERGENCY HAZARD SIMULATION TESTS

- **Heatstroke Scenario:** Skin temp (42°C) + low HRV triggers CRITICAL status (73% Anomaly Risk) & 880Hz local siren.
- **Toxic Gas Leak:** VOC gas surge (58 PPM) + SpO2 drop (91%) triggers instant offline LoRa distress broadcast packet.

PROTOTYPE EVIDENCE 05: DGMS STATUTORY AUDIT & AIR-GAPPED OT SECURITY



IN VITALSGRID SCADA OT | INDIAN INDUSTRIAL COMPLIANCE

OFFICIAL INDUSTRIAL INCIDENT & WORKER SAFETY AUDIT REPORT

Statutory Standards: DGMS Coal Mines Reg 2017 | The Factories Act, 1948 (Sec 41B) | IS 14489 | CEA Safety Regs 2010

REPORT ID: DGMS-INC-2026-523072
TIMESTAMP: 3/9/2026, 12:14:34 pm IST
SECURITY LEVEL: AIR-GAPPED OT (ZERO WAN)

1. WORKER & PLANT FACILITY IDENTIFICATION

Worker Badge / Aadhaar ID: **WORKER_001 (IND-EMP-9972)**
Assigned Plant Zone: **Control Room**
Facility & Location: **SITE_A**
Wearable Node ID: **EDGE_A1**
Industrial Work Shift: Shift A (06:00 - 14:00 Morning)
Emergency Dispatch: **+91 112 / Plant Hospital Medical Response**

2. ON-DEVICE TINYML ANOMALY EVALUATION

Risk Classification: NORMAL EMERGENCY
Anomaly Score: **1.0%**
TinyML Engine: **VG-TinyML-Fusion-v0.3**
MCU Latency: **8.1 ms (On-Device nRF5340)**

3. CRITICAL TELEMETRY SNAPSHOT AT TIME OF INCIDENT

Sensor Module	Parameter	Recorded Value	DGMS / Factories Act 1948 Limit	Evaluation
TI AFE4900 ECG	Heart Rate / HRV	110 BPM / 45.0 ms	HR < 120 BPM / HRV > 35 ms	PASS
ST LSM6DSOX IMU	Motion Jerk / Fall Impact	0.90 m/s³	Jerk < 3.0 m/s³	PASS
Sensirion SGP40	Toxic SF6 / Gas Leak	68 PPM	Gas < 35 PPM (DGMS Norm)	TOXIC LEAK
Sensirion HTS221	Ambient Heat Index	28.5 °C	Heat < 37.5 °C (Indian Summer Limit)	PASS

4. AIR-GAPPED DISPATCH LOG & INDIAN REGULATORY DISPATCH

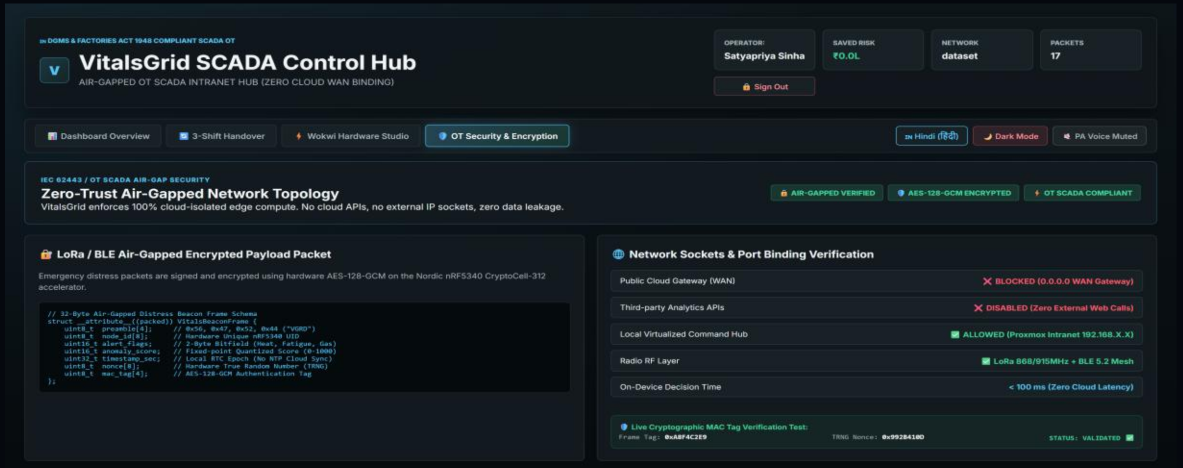
On-device haptic buzzer activated in **<100ms**. Offline LoRa/BLE distress beacon broadcasted to SCADA Intranet Command Hub without cloud dependencies. Emergency protocol alert logged for DGMS Regional Safety Inspector audit under Factories Act, 1948 Section 41B.

SCADA Site Chief Supervisor (Plant Head)

DGMS / Factory Inspector Safety Auditor

Print / Save PDF | Export JSON | Export TXT Log | Close Report

0.3



STATUTORY DGMS AUDIT & OT SECURITY

- **DGMS Audit Form IV:** Mines Act 1952 statutory incident investigation document for official audit logging.
- **AES-256 Mesh Encryption:** Zero cloud WAN telemetry leakage; local mesh payload encrypted end-to-end.

HACKCULTURE 2026 // TECHNICAL PROPOSAL SUBMISSION

READY FOR THE EDGE.

Air-Gapped Edge AI Wearables for Occupational Health in Critical Infrastructure

ARCHITECTURE SUMMARY & IMPACT

- **Cloud-Independent SCADA Hub:** 100% air-gapped intranet operation with zero external REST WAN binding.
- **Deterministic Edge AI:** Sub-50ms TinyML sensor fusion on Nordic nRF5340 dual-core MCU.
- **Statutory Audit Compliance:** Mines Act 1952 & Factories Act 1948 compliant DGMS Audit Form IV generation.
- **Instant Distress Response:** Local 880Hz audio alarm, haptic motor, and offline BLE/LoRa mesh beacon.

ARCHITECTURE TEAM

1. **Abhiraj Raushan**

   Lead Architect & Developer

2. **Satyapriya Sinha**

   Co-Lead & Systems Engineer

Thank you for reviewing the VITALSGRID technical proposal.